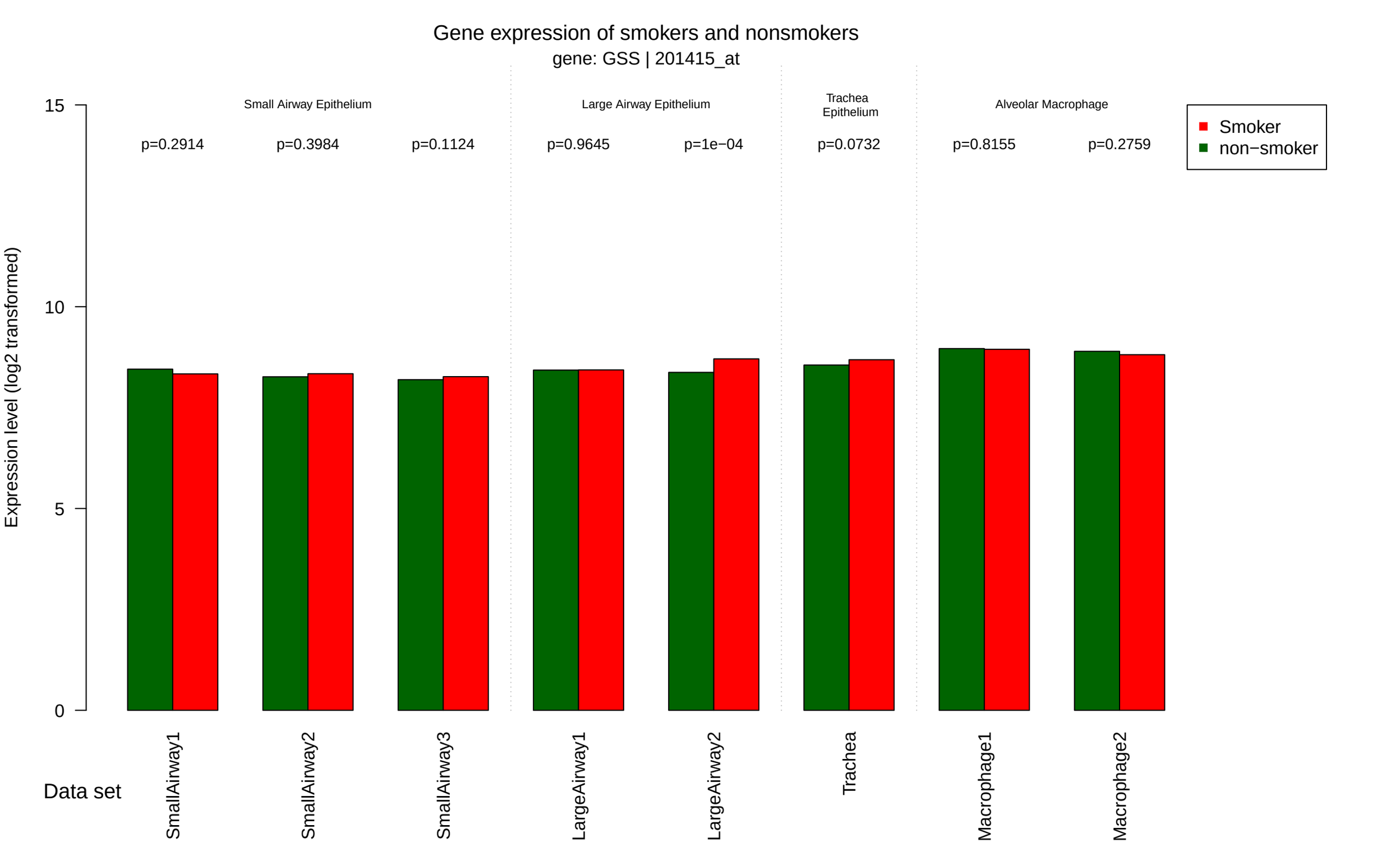
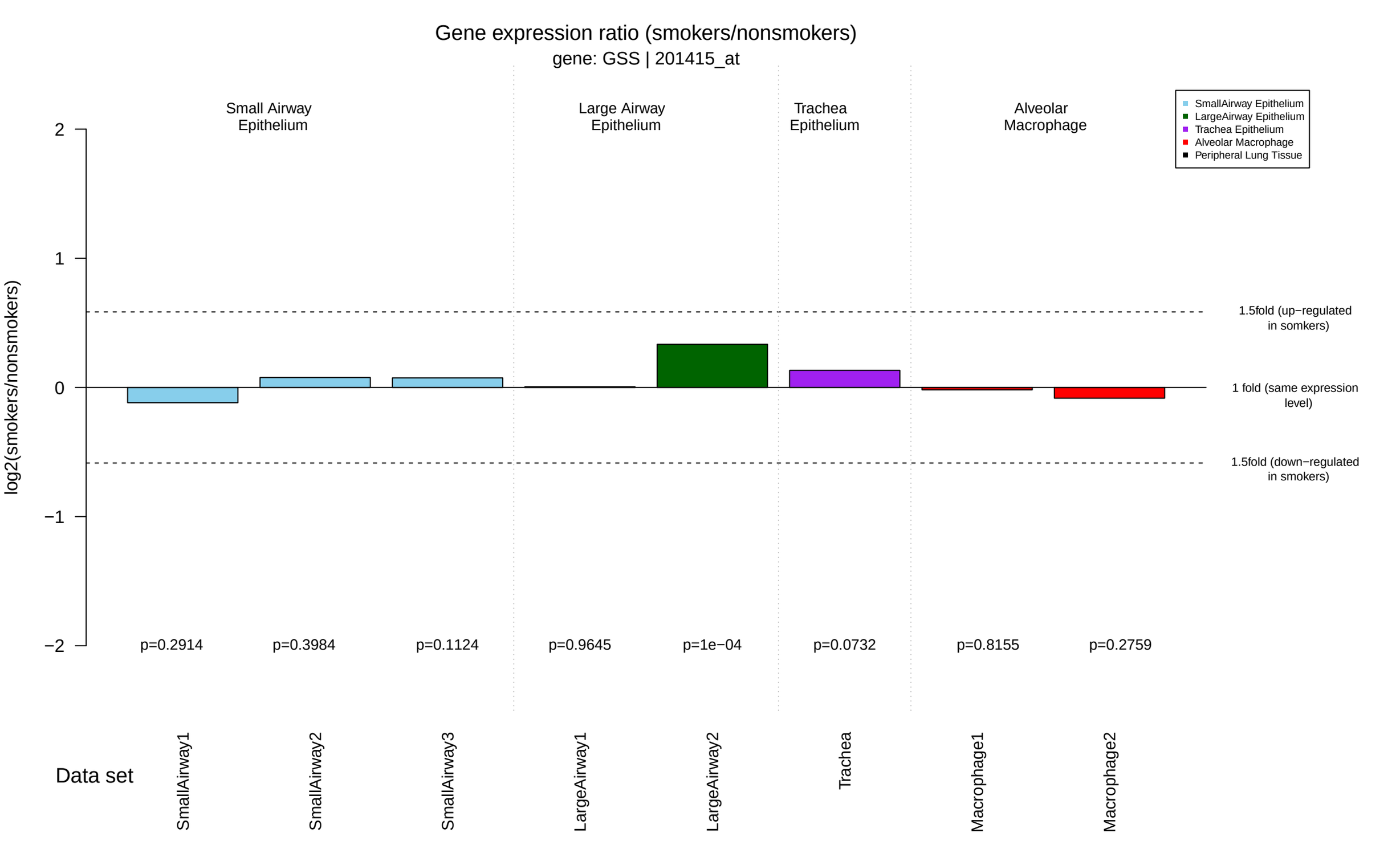
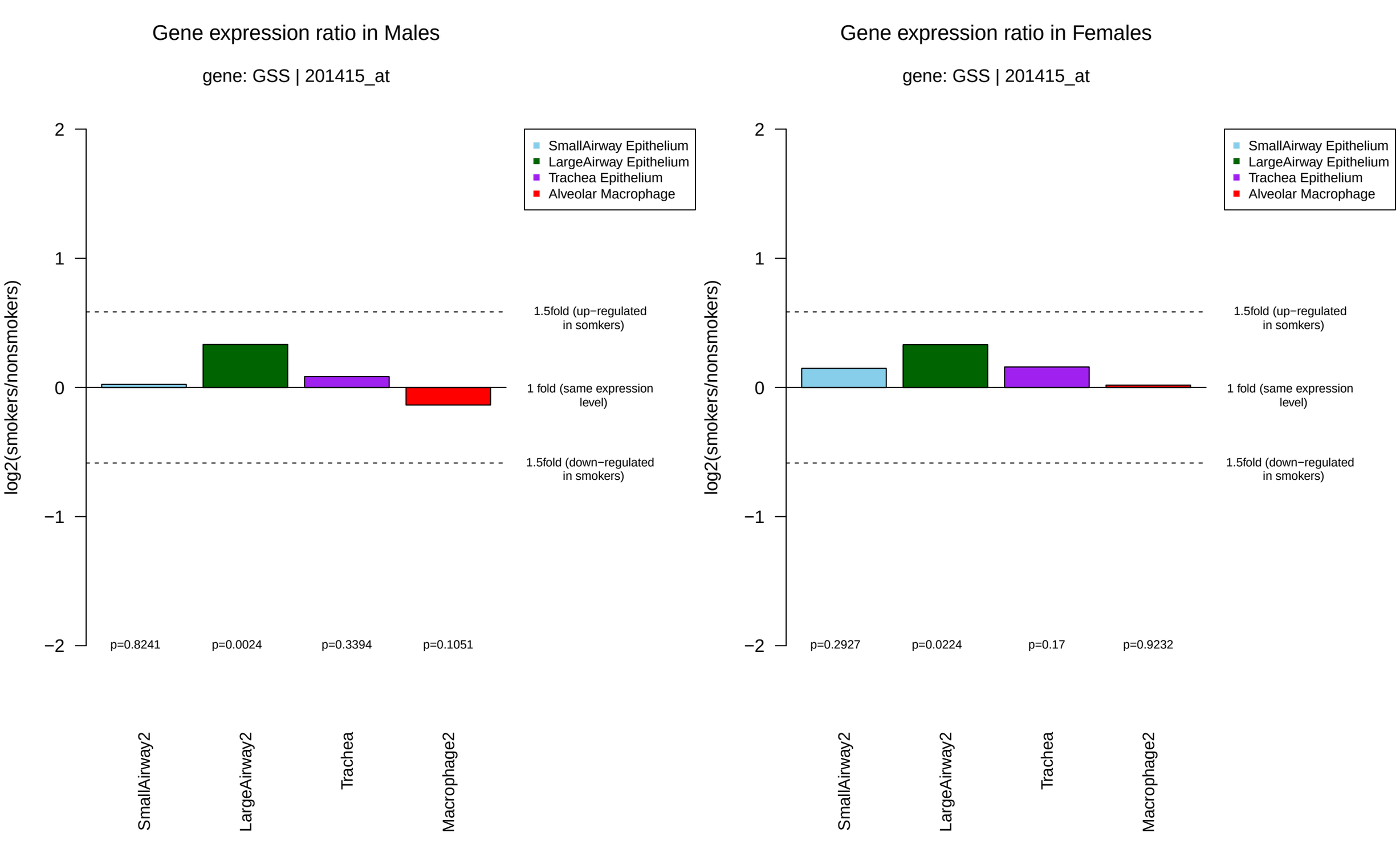
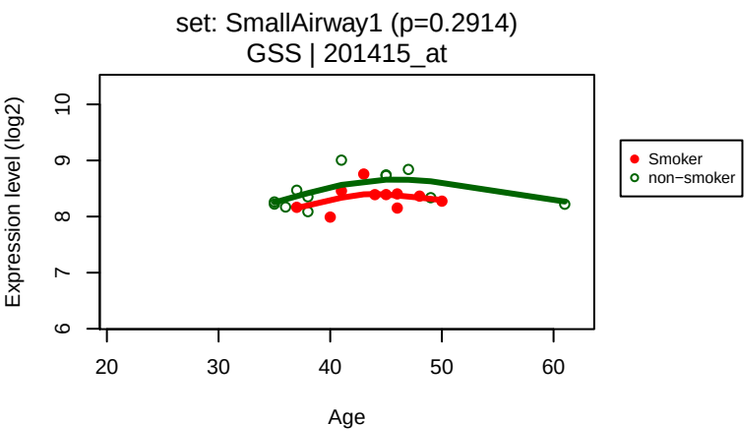
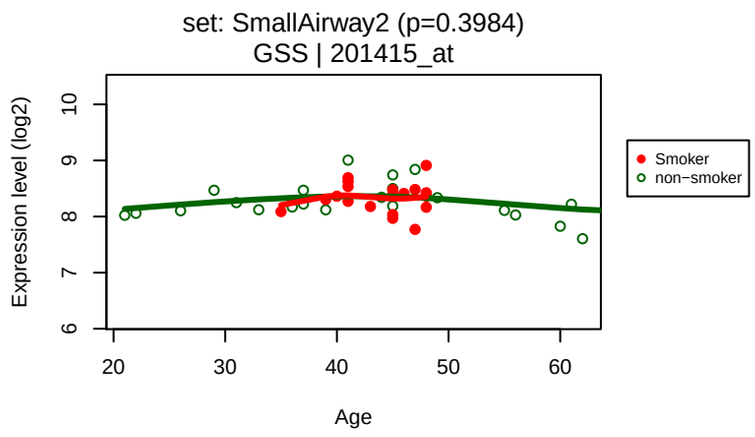
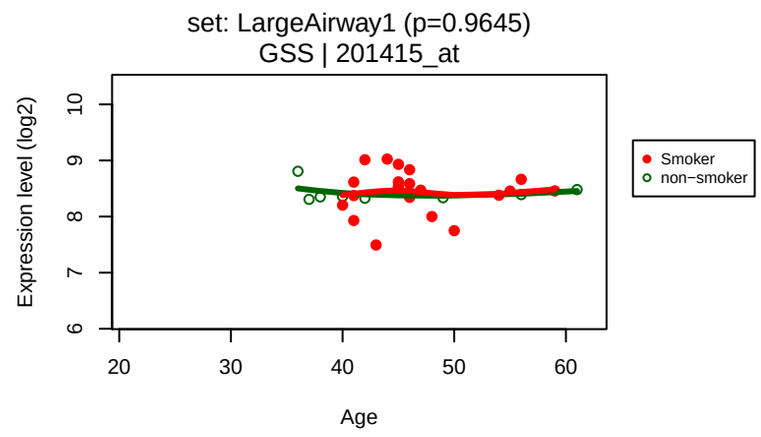

Gene expression of smokers and nonsmokers
gene: GSS | 201415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2914
p=0.3984
p=0.1124
p=0.9645
p=1e−04
p=0.0732
p=0.8155
p=0.2759
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

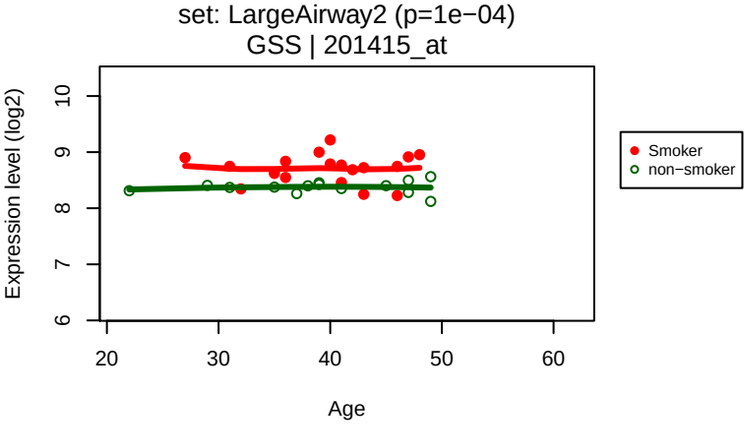

Gene expression ratio (smokers/nonsmokers)
gene: GSS | 201415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2914
p=0.3984
p=0.1124
p=0.9645
p=1e−04
p=0.0732
p=0.8155
p=0.2759
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

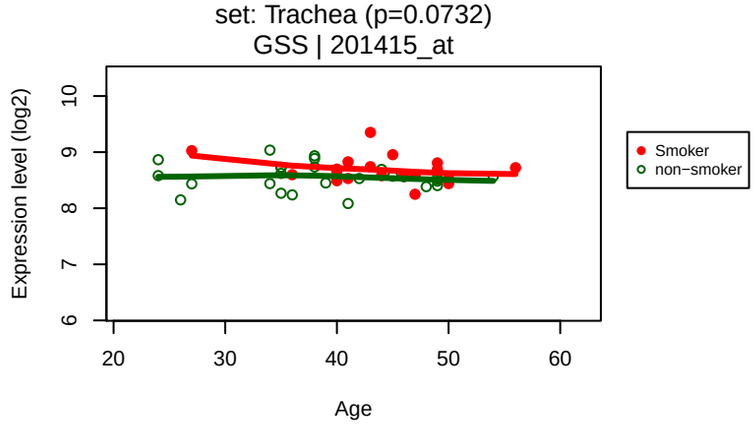

Gene expression ratio in Males
gene: GSS | 201415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8241
p=0.0024
p=0.3394
p=0.1051
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSS | 201415_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2927
p=0.0224
p=0.17
p=0.9232
SmallAirway2
LargeAirway2
Trachea
Macrophage2

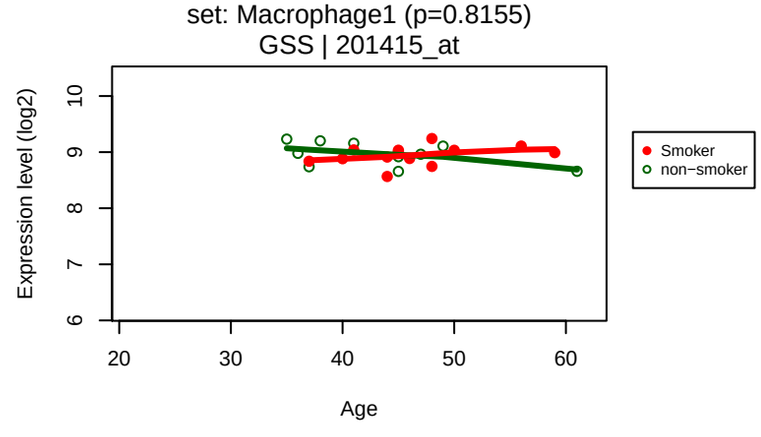

set: SmallAirway1 (p=0.2914)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker

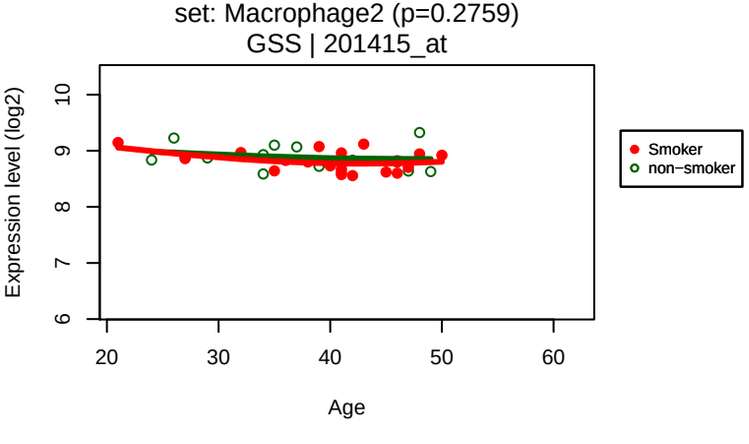

set: SmallAirway2 (p=0.3984)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker

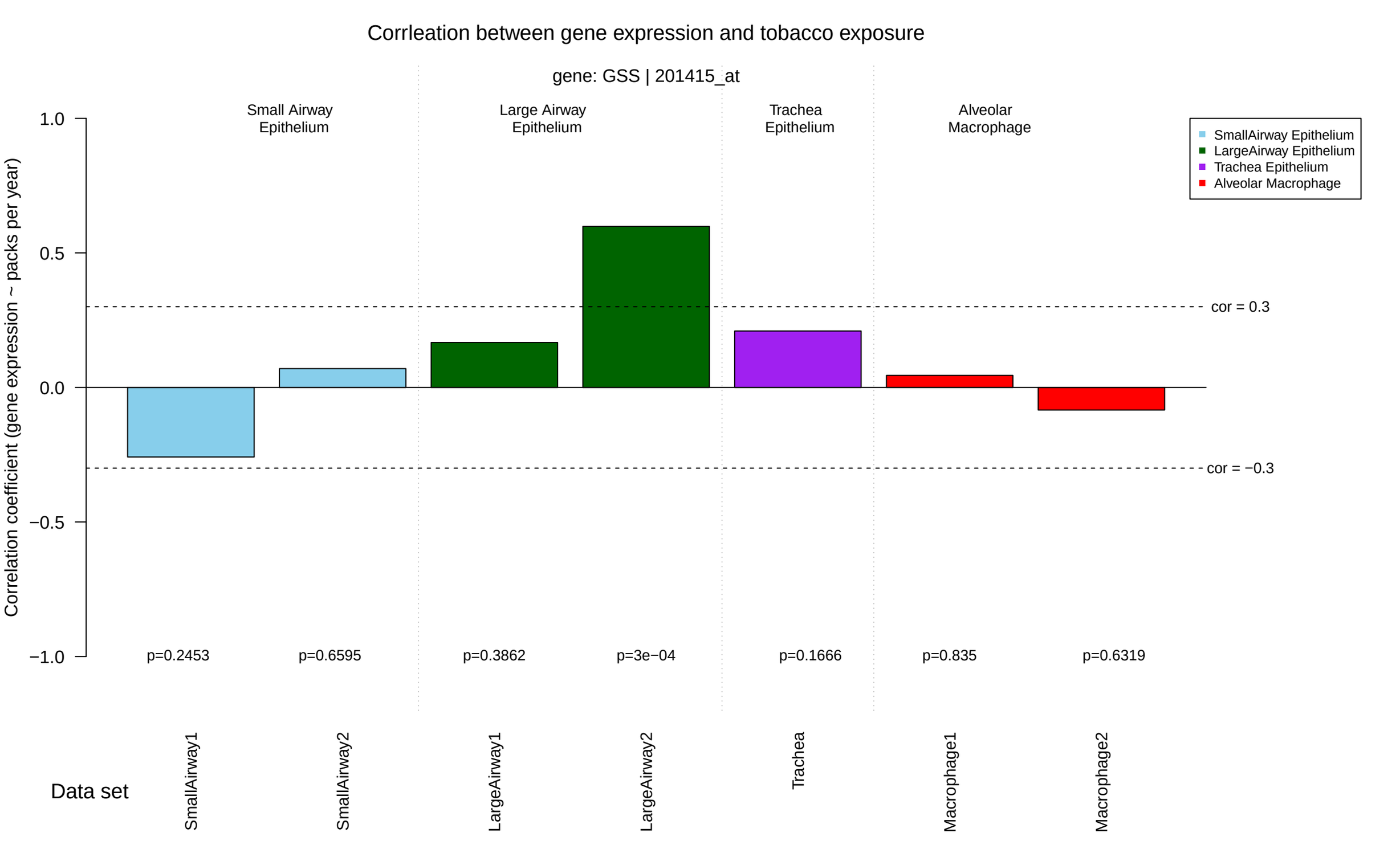

set: LargeAirway1 (p=0.9645)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=1e−04)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0732)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8155)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2759)
GSS | 201415_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSS | 201415_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2453
p=0.6595
p=0.3862
p=3e−04
p=0.1666
p=0.835
p=0.6319
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2